

USER GUIDE / EDITION 08 OCTOBER 2026

EUROFIGHTER IFE FUEL CALCULATOR

Operating guide and capability overview

Version 1.9.19 | Simulator mission planning

Purpose

Plan a mission from initial inputs to a printable Data Card: route, timing, fuel, holds, AAR, diverts and fuel thresholds. Save the complete current plan as JSON and restore it on another device.

Capabilities at a glance

AREA	CAPABILITY	P.
MISSION / ROUTE	Mission details, SimBrief import, waypoint editor and vertical route profile.	02
FUEL / CRUISE	Forward calculation from entered start fuel; separate cruise calculator.	03
TIMING / RECOVERY	Hard times, holds, WP IN/OUT, Joker, Bingo and two diverts.	04
AAR	AUTO/MANUAL, gross onload, transfer rate, transfer time and event burn.	05
BACKUP / RESTORE	Complete JSON export/import and local mission storage.	06
OUTPUT / INTERFACE	Data Card, AAR ticket, tablet controls, language and appearance.	07
WORKFLOW / LIMITS	Planning sequence, model limitations, validation and terminology.	08
ILLUSTRATED GUIDE	Configure AAR, select diverts and read the AAR ticket.	09

Document scope: a practical product and operating overview. This is not an official flight document or military regulation. It describes implemented features in this software version, not planned extensions.

Mission and route

Create, import and edit a plan while retaining control of its inputs.

Mission setup

Enter Mission Name, Mission Type, UTC date, Start Z, Aircraft, Callsign, Departure/Arrival, airfield elevations and mission notes. Start fuel, taxi time, reserve and a fuel consumption surcharge are editable. Airfield elevations can be taken from the local database or entered manually.

ENTRY POINT	BEHAVIOUR
NEW MANUAL MISSION	Start a manual plan with editable intermediate waypoints.
LOAD DEMO	Load the existing demonstration route with DIST/TIME legs and two divers. The app still starts with the demo.
SIMBRIEF	Import the latest OFP using a Pilot ID or alias. Imported data include date, route, coordinates, altitudes, Mach and wind. Planning values remain editable afterwards.
LOAD PRESET	Explicitly load a separately offered mission preset. Selecting a preset in the list alone does not change the active mission.

Route / leg editor

Fields are ordered **WP - ALT ft - Mach - time/ETA Z - NM - Hold/WP OUT**, followed by wind, mode, manual leg minutes, ETE, leg fuel, recovery requirement, FOB and actions. Coordinates remain part of the internal route state and backup.

Intermediate waypoints can be added, deleted or moved up/down. AAR, Bingo and Joker references follow their associated waypoint when it moves. Deliberately reordering a route may recalculate affected DIST legs from available coordinates.

Distance or time: choose the governing input

DIST: NM is specified; ETE follows from speed, wind and altitude changes. **TIME:** entered leg minutes govern the calculation; leg distance is derived from them. Editing NM or MIN selects the corresponding mode. Holds, AAR duration and hard times do not change leg NM or coordinates.

Precision and editing

Mach is displayed with no more than two decimal places. Full precision is retained internally. Route edits commit on Enter or when leaving the field. Leaving a rounded field unchanged does not overwrite its precise internal value.

Fuel and performance

Separate the planned fuel state from the required minimum.

Start / On Board Fuel is an input. The app calculates fuel states forwards from this value. A desired landing quantity or AAR target does not back-calculate and overwrite start fuel.

DISPLAY	MEANING
FOB START / ON BOARD	Entered fuel on board before the planned ground consumption.
BLOCK REQ	Planning requirement including route, holds, reserve, divert/recovery and the net AAR effect.
T/O REQ	Required take-off fuel; this is not the actual or planned tank content.
PLANNED T/O FOB	Planned fuel on board after subtracting ground consumption from start fuel.
UPLIFT / SPARE	Additional fuel needed to meet the block requirement, or the difference between on-board fuel and the planning requirement.
FOB IN / OUT / LANDING	Calculated fuel states at waypoints, after events and at landing.

Calculation basis

Climb/descent and cruise portions are calculated separately for each leg. Mach is converted to TAS using ISA; the entered wind component affects groundspeed. The fuel curve interpolates between measured points and uses the stored estimation model outside the measured Mach range.

The supplied EF2000 profile contains tank capacity and ground, climb and descent rates. The active Mach range is 0.48 to 0.95; model altitude extends to 55,000 ft. The airborne consumption surcharge applies to flight phases including holds and AAR burn. Ground consumption is calculated separately from taxi minutes and the ground rate.

Separate cruise calculator

Calculate by total distance or total flight time. Distance may come from the current route, the direct DEP-ARR distance or manual entry. Set cruise altitude, Mach, wind and both airfield elevations. Outputs include climb, cruise, descent and total fuel, fuel flow, ETE, distance and block requirement including reserve and recovery/divert.

Legs too short for an altitude change, non-positive groundspeed and out-of-model values are flagged for review. Missing performance data are not filled by inventing a new aircraft profile.

Timing, holds and recovery

Time constraints and position-dependent fuel thresholds.

Hard times and holds

Start time and ETA use Zulu/UTC. A manual waypoint ETA acts as a hard-time anchor. Additional waiting time is treated as a hold with fuel consumption, preferably at a planned hold waypoint before the anchor. Unreachable times are flagged as conflicts; distances are not invented to make them fit.

A hold changes **time and fuel**. Each affected waypoint has separate WP IN time/FOB, hold duration/burn and WP OUT time/FOB. Consumption uses the hold waypoint altitude and Mach plus the airborne consumption surcharge. Subsequent ETAs shift accordingly.

Joker and Bingo

FUNCTION	LOGIC
JOKER @ WP IN	The hold at the selected waypoint is still included in the remaining requirement.
JOKER @ WP OUT	The hold at the selected waypoint is complete and is not counted again.
JOKER BUFFER	An editable additional buffer above the calculated requirement at the selected reference point.
BINGO REFERENCE	Position-dependent requirement for the remaining mission plus divert, recovery extra and landing reserve. At arrival, remaining mission fuel may be zero.
MANUAL / AUTO	Bingo and Joker can be calculated or replaced with explicit manual thresholds. The app shows the reference point, planned FOB and margin.
BINGO 2 / B2 RTB	A separate entered threshold with its own RTB airfield on the AAR plan.

Divert and reserve

Plan two alternate airfields. Search supports ICAO, name, country, favourites and recent airfields. Distance and elevation may use local data or manual entries. Divert altitude, Mach, wind, Missed Approach / Recovery Extra and Landing Reserve remain editable.

USED DIVERT can automatically select the higher valid fuel requirement or explicitly use Divert 1/2. A preset may contain labelled briefing fuel values. AAR onload that has not yet been received is not credited as fuel already available when determining conservative remaining fuel thresholds.

AAR: fuel, transfer and event

Gross onload, receiver burn and time requirements are shown separately.

Enable AAR at an intermediate waypoint. Inputs include ARIP, ARCP, planned ARCT/OUT, FL, KIAS, configuration, tanker type, callsign, frequency, AUTO target or manual gross onload. **Transfer Rate and Event Duration remain independent inputs.**

QUANTITY	USE
TRANSFER RATE	Effective gross onload in kg/min. Manually editable; no verified real-world rate is supplied.
RATE SOURCE	MANUAL. The backup format also supports TANKER PRESET / RECEIVER PRESET provenance. No pre-filled verified rate presets are included.
TRANSFER TIME	Time during which fuel is actually transferred. Without a rate: N/A / RATE REQUIRED.
AAR EVENT DURATION	Total planned event time. A longer event includes additional time with normal AAR/orbit consumption.

Calculation

Notation: R = gross transfer rate; F = receiver fuel flow; G = gross onload; D = event duration; t = transfer time. Fuel in kg, time in min, R/F in kg/min.

MANUAL: $t = G / R$

AUTO, transfer only: $t = \max(0, \text{Target} - \text{IN}) / (R - F)$

AUTO, longer event: $G = \max(0, \text{Target} - \text{IN} + F \times D)$

$\text{TransferBurn} = F \times t$; $\text{OrbitBurn} = F \times (D - t)$

$\text{FOB OUT} = \text{FOB IN} + G - \text{TransferBurn} - \text{OrbitBurn}$

For AUTO with a longer event, $t = G/R$. TARGET FOB OUT always means the fuel state at the end of the entire event. These formulas require a valid plan with $R > F$ and $D \geq t$.

Safeguards and ticket

$R \leq F$ produces no valid transfer time. An event that is too short is flagged EVENT TOO SHORT and is not extended automatically. Model sequence: **transfer first, orbit afterwards**. IN, peak tank content at transfer end and OUT are checked. FOB AFTER LOAD is an accounting value only.

The ticket and Data Card show FOB IN, gross onload, rate, source, EST XFER TIME, event duration, AAR burn, FOB OUT and AAR OUT. Without a rate, existing fuel planning based on planned event burn remains visible; transfer time and transfer feasibility are explicitly unverified.

Complete mission backup

Plan once, save to a file and restore on another device.

EXPORT CURRENT MISSION / IMPORT MISSION

Both controls are under MISSION. Export creates a JSON file of the current state; SAVE LOCAL is not required first. Import validates the file, restores the inputs and recalculates the plan. A successful import displays **Mission successfully imported**.

SAVED	COVERAGE
Mission inputs	Name/type, date, start, aircraft, callsign, DEP/ARR, elevations, notes and all calculation-relevant form values.
Internal route state	Order, names, available coordinates, ALT, exact Mach values, wind, NM, modes, manual times, hard times, planned times and holds.
Events and thresholds	AAR switch/reference and all AAR inputs including rate/source; Bingo/Joker switches, WP references, IN/OUT basis and manual values.
Recovery / performance	Diverts, reserve, manual distances/elevations, cruise settings, metadata and the aircraft profile, fuel curve and navigation data in use.
Results	A reference snapshot of leg, fuel, AAR and summary values. It never replaces recalculation on import.

Version handling

Current format: **schemaVersion 2**, appVersion 1.9.19 and calculation model

ife-forward-transfer-2. Schema 1 is migrated explicitly: only the previously absent rate fields are added, with a blank rate and MANUAL source. Unknown versions/models or missing required fields are rejected before loading.

File limit: 5 MB. The complete format supports 2 to 2,000 waypoints and does not truncate a backup route to the earlier 40-WP limit. Blank inputs stay blank; unknown coordinates are not estimated.

Local storage

SAVE LOCAL and LOAD LOCAL save/restore on the same device. New local saves use the same complete structure as JSON export; older browser saves remain readable through the existing loader. Export, file import and preset loading do not overwrite the saved local mission. Deliberate SAVE LOCAL replaces the local storage slot.

JSON is the authoritative complete backup format. PDF is for reading, not reimporting. An additional CSV/Excel export is not implemented in this version.

Output and interface

Cockpit-style information with display precision separate from calculation precision.

Data Card and AAR ticket

The Data Card combines the mission header, fuel order, Bingo/Joker, diverts/reserve and AAR. The flight log includes WP, ALT, Mach, wind, leg NM, ETE, WP IN, Hold/WP OUT, leg fuel, recovery requirement and FOB IN/OUT. An actual-fuel column (IST kg) is provided for recording observations.

PRINT / SAVE PDF uses the browser print dialogue. Print the complete Data Card or the separate AAR ticket. Use the JSON backup for full restoration.

Tablet controls and detail views

On touch devices or narrower screens, route rows become expandable waypoint cards. Tap the full waypoint name to open or close the fields. The desktop table remains available, with a waypoint column that stays visible during horizontal scrolling.

The vertical route profile shows the altitude sequence and marks AAR. Selecting a waypoint in the profile opens leg, timing, consumption and fuel-state details. Key fuel displays also provide detail views.

AREA	FUNCTION
LANGUAGE	English is the default. Settings → Language switches between English and German. Previously saved choices remain effective; times use UTC.
LAYOUT	OPS and OPERATIONAL EFB display variants.
THEMES	Night Cyan, Night White, Ops Green and Day High Contrast.
LOCAL OPERATION	Browser-based app with a service-worker cache. Calculations run locally; offline use requires previously loaded/available app files.
DATA STATUS	Profile/fuel data, AAR measurement status and AIRAC calendar cycle. The cycle does not certify that navigation data are current or approved.
PRIVACY	No app account or tracking SDKs. SimBrief is contacted on explicit import. Local app data can be deleted after confirmation.

Optional mission preset

RHINE SENTINELLE - GHOST - 10.10.2026 can be loaded separately. It contains the stored route, timing, holds, AAR and briefing fuel values. It remains optional and is not the app default. An unknown transfer rate is not filled in for this preset either.

Workflow and limitations

Use the simulator model with its assumptions in view.

SEQUENCE	CHECK
01 / CREATE	Choose a manual mission, SimBrief or an optional preset. Check date, start, airfields and start fuel.
02 / ROUTE	Check order, coordinates, altitudes, Mach, wind and DIST/TIME modes.
03 / TIMING	Set hard times and holds. Check WP IN/OUT and timing conflicts.
04 / RECOVERY	Set divers, Used Divert, Recovery Extra, reserve and Joker/Bingo references.
05 / AAR	Check AUTO/MANUAL, gross/target, configuration, rate and event duration. Resolve time and tank warnings.
06 / SAVE	Review recalculated results, export JSON and print the Data Card/AAR ticket to PDF if needed.

Model limitations

The calculator uses simulator measurements, interpolation and identified model assumptions. Below 3,000 ft it uses the 3,000-ft curve; a separate measured low-level dataset for every altitude is not available. Weather, weight and configuration are not covered by a complete real-world performance database.

The active AAR consumption basis is the existing 3-BAGS profile. Archived CLEAN AAR runs are provisional and do not provide an active verified gross-transfer curve. Climb/descent rates and KIAS-to-Mach conversion are model assumptions. CHECK ROUTE, AAR PROFILE, RATE REQUIRED, EVENT TOO SHORT, CAP LIMIT and HARD TIME CHECK call for review of the inputs.

Validation and document basis

For v1.9.18, 15 existing regression checks, 10 backup test groups, 9 new AAR transfer test groups and route-input tests were run. Coverage includes export-reset-import, precise restoration, AUTO/MANUAL, peak tank content, event times and unchanged geometry. Synthetic test rates are not real-world calibration. These checks do not establish operation on a particular tablet or approval for actual flight.

Basis: app package v1.9.19; app.js, logic-core.js, mission-backup.js, index.html, data files, README and related regression tests. This guide describes the implemented scope as of 8 October 2026.

Terminology

FOB Fuel on Board · **ETE** Estimated Time Enroute · **ETA** Estimated Time of Arrival · **AAR** Air-to-Air Refuelling · **WP** Waypoint · **NM** Nautical Miles · **RTB** Return to Base · **TOT** Time on Target

SIMULATION ONLY. Not an official flight document or approved for real-world operational planning.

Set up AAR and divers

Illustrated operating guidance with the current English field names.

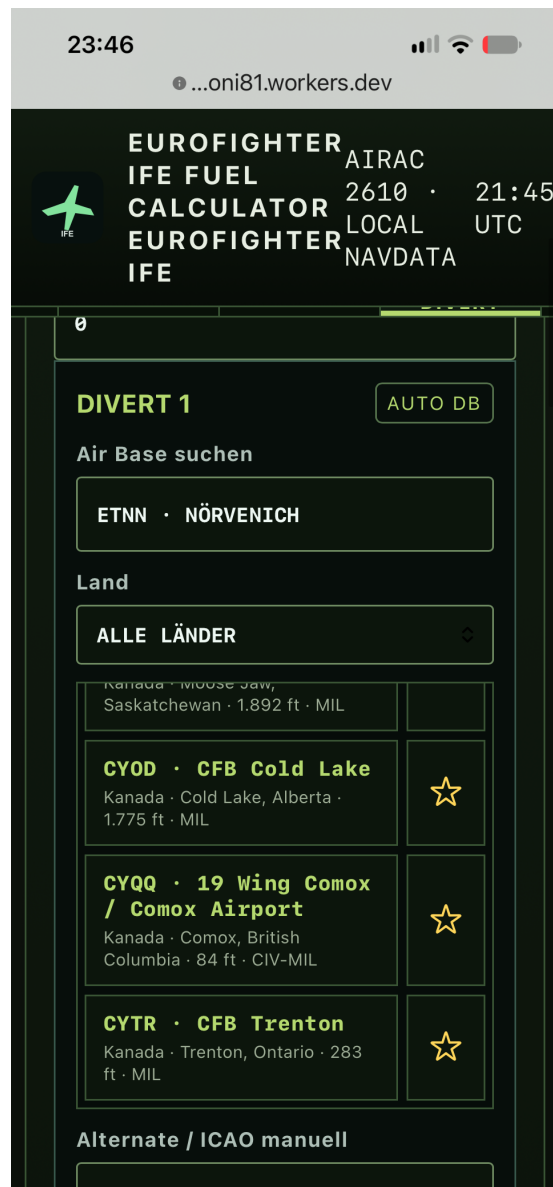
These supplied screenshots show earlier interface versions and retain their original language. They illustrate where controls are located; their mission values are not defaults for a new plan. Follow the current fields and explanations in this guide.



01 / ENABLE AAR

In the **AAR** tab, enable **PLAN AAR**. Select the associated waypoint and check tank capacity. Enter ARIP, ARCP, altitude, KIAS and configuration. Then set the AUTO target or manual gross onload, and event duration.

Image sources: supplied screenshots IMG_3169(1).png and IMG_3170(1).png. Appearance and selected values reflect the respective capture states.

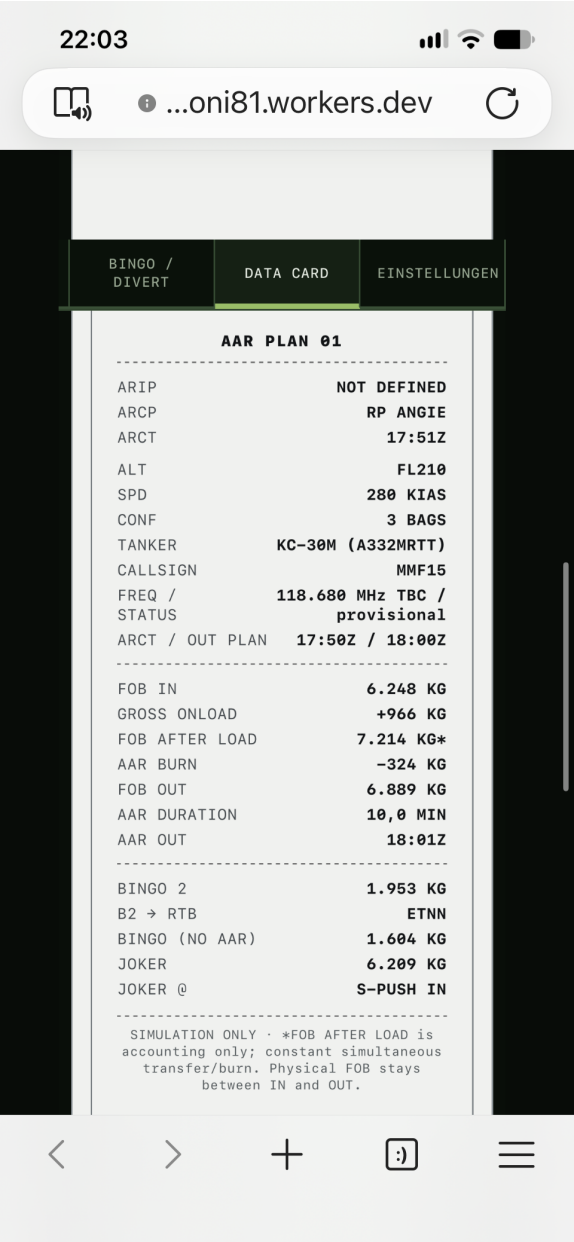


02 / SELECT AN ALTERNATE

Under **DIVERT 1** or **DIVERT 2**, search by ICAO or name. Optionally filter by country; the star saves a favourite. Select the airfield, check its values and set **USED DIVERT**. Review reserve and Recovery Extra separately.

Read the AAR ticket

Distinguish planned times, calculated FOB and unavailable transfer rates.



03 / FUEL SEQUENCE

FOB IN is calculated fuel before the event. **GROSS ONLOAD** is fuel transferred. **AAR BURN** is event consumption. These determine **FOB OUT**.

04 / CURRENT VERSION

The older screenshot has no transfer-rate lines. The current ticket adds **TRANSFER RATE, EST XFER TIME** and **AAR EVENT** between gross onload and AAR burn.

05 / RATE REQUIRED

Without a confirmed Effective Transfer Rate, rate and transfer time remain **N/A / RATE REQUIRED**. No additional fuel or timing values are inferred from a missing rate. Consumption for the independently planned event duration remains in the plan.

06 / PLAN AND RESULT

ARCT / OUT PLAN are planned times. Calculated ARCT and AAR OUT may differ. Review discrepancies; this screenshot does not establish a currently valid mission plan.

07 / ACCOUNTING

FOB AFTER LOAD is an accounting intermediate. With a confirmed rate, the current model checks transfer first, orbit afterwards and actual peak tank content. The older image footer does not describe this current sequence.

Save the plan

After review, select **EXPORT CURRENT MISSION**. The JSON file saves the complete plan. **PRINT / SAVE PDF** produces a readable Data Card or AAR ticket; this printout does not replace the JSON backup.

Image source: supplied screenshot IMG_3214.png, an earlier AAR ticket. The other supplied images were reviewed to identify differences from the earlier interface.